

Chapter 9

Innovating for Curricular Complexity: Executive Education for Transdisciplinary Challenges



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9.1 Introduction

Executive education amplifies university-partner collaborations and can be effective in initializing new relationships. Executive education student cohorts, whether attending a standing certification program or an *ad hoc* workshop, are vectors through which university faculty and administration can engage with external partners. Leveraging existing curricula and course syllabi from existing degree programs as the basis for executive education program design inaugurates a discussion with partners about the relevance of current university degree programs to workforce needs, particularly at the graduate level. This accrues institutional benefits as course instructors strive to keep their syllabi focused on the needs of the workforce marketplace and society at large, and may provide important feedback to the basic and applied research enterprise. A well-organized and topically relevant executive education program affords university faculty opportunities to explore common interests with external partners, strengthening engagements and producing new and perhaps co-produced scholarship.

Institutional perception and priorities as well as an understanding of market needs play large roles in the design of executive education programs. Often, executive education curricula are based on what is being taught in existing graduate degree programs in the professional schools. The learning objectives in these

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programs can be mapped into bespoke market- or partner-sensitive outcomes, which, in turn, can lead to adjusting the fit of degree program syllabi to concentrated executive education intensives. This mapping exercise makes use of the efforts already expended on the degree programs, while making the executive education program more accessible to professionals who may not be able to afford the time needed to engage with a semester-long course, let alone a multi-year degree commitment.

Faculty commitments to executive education may follow the “one instructor per course” model, with perhaps a handful of additional instructors contributing specialized material or participating in panels and giving keynotes. While some courses clearly can be innovative takes on contemporaneous market needs and partner expectations, as a whole, executive education course lists tend to conform to the topical granularity present in the degree programs.

This symbiosis between executive education and university degree programs is challenged when university initiatives in research and degree education, or when the interests of the partners supplying student cohorts, take sharply divergent trajectories. Driven by emerging or disruptive scholarship, or by emerging and disruptive trends in society affecting business operations, or by both, this dispersion requires innovative curricular design, new executive education program structure and faculty engagement models, and enhanced and aggressive discussions with external partners about mutual interests. Among these interests are the scholarly and operational issues inherent in the increasingly apparent rise of complex, multidisciplinary challenges faced broadly by human society, along with the imperative to identify practicable solutions. We use the term “transdisciplinary” to describe multidisciplinary problems having complex and granular interdependencies among the disciplines.

Many universities are recognizing a role beyond the development of new knowledge and evolution of disciplinary degree programs. As scholarship becomes more transdisciplinary in response to the need to understand complex human and natural systems, the architecture of university research and education is beginning to cross or deconstruct disciplinary silos, with new institutes and professional schools developing to accommodate these new knowledge trajectories. Cross-disciplinary collaboration and new intellectual threads are leading to new and innovative university narratives about the university’s role in relating the production of knowledge to solutions needed for society’s most complex challenges.

The evolution of complex, transdisciplinary thinking in response to these challenges—what some have characterized as advanced system thinking or convergence research—is not just confined to university campuses. The public and private sectors, nonprofits, nongovernmental organizations, and other enterprises are facing a combination of client and stakeholder demands, new regulatory environments, new business initiatives and a changing entrepreneurial landscape, as these complex challenges begin to have a socioeconomic impact and force a focus on enterprise response. This is happening rapidly, for example, with regard to challenges such as climate change, global pandemics, cybersecurity, the explosion of data of all types and rapid developments in artificial intelligence.

This new and emerging research and educational environment motivates innovation in executive education, even as its effects on the structure of university research, degree programs and even departmental and school administration are not yet fully evident.

We explore here the kinds of innovations needed to develop executive education curricula for overly complex topics with intersections across a wide range of disciplines. We aim to understand how complex topics—the kind that are reflected in the evolution of the university itself—can be mapped into short intensives that accelerate an education partner’s adaptation and response to new operational landscapes.

9.2 Challenges and Opportunities: Open Enrollment vs Problem-Oriented Cohorts

The bulk of executive education programs are designed based on a market need, as understood by program leaders and their faculties. Within the academy, the design might rely on the curricula developed for the degree programs in professional schools such as law and business, or through interactions with external partners in government, the private sector, nongovernmental organizations, professional societies, and philanthropies. Some design work is reinforced by market analyses of workforce needs for upskilling, certifications and accreditations, or by new legal or regulatory externalities affecting the operations of university partners.

Open enrollment executive education, typically based on per-seat fees, engages student cohorts with no particular commonalities other than a desire to learn, not a small attribute. Of course, such a program serves a general educational purpose and signifies a university’s aspiration to engage with lifelong learners. At the same time, executive education exists at the interface with industry and other external partners, and this interface can have a more strategic role in developing and stewarding university-industry relationships. Programs at this interface reinforce commitments to external at-large stakeholders. Perhaps more importantly, well-designed bespoke programs can be effective at translating core research and educational innovation to real-world practice, generating new industry partnerships and strengthening existing relationships. This works particularly well when new university scholarship and educational initiatives align with the strategic objectives of external partners. When either outstrips the other, there will be challenges.

Such challenges arise when scholarship and educational pedagogies begin to study and address transdisciplinary problems, demanding innovative collaborations and administrative structures that help build new research and program trajectories. Typically steered by faculty polymaths and innovative university leaders, these initiatives can come to symbolize university growth and engagement with the broad challenges facing greater society.

Examples abound. Artificial intelligence, public health, globalization, national security, and climate change manifest as a combination of complex technical and socioeconomic challenges, new multidisciplinary research into basic processes, strategic and operational demands on the public and private sectors, geopolitical

responses, and university and industrial R&D pursuing solutions. The challenges this complexity imposes on university executive education requires innovations in individual syllabi and total curriculum development, delivery modes and pedagogy, marketing, and, importantly, management and stewardship of university-industry and other external partners.

The need for innovation in executive education arises when faculty research, pedagogic innovation and new university initiatives outstrip existing executive education curricula, or when new externalities such as policy and regulatory shifts change the demand side of the programs. Some of the most challenging drivers of change occur when student and partner interest in complex societal challenges create new curricular demands. Our experience with one such driver, climate change and sustainability, serves as an example from which more general observations about executive education can be drawn. We first present general concepts, and will illustrate their use in a climate-focused case study.

9.2.1 Generalized Curricular Design for Complex Subjects

We delineate four categories requiring innovation. First, the subject itself is transdisciplinary; that is, its scholarly underpinnings are, at a minimum, multidisciplinary, intersectional, and evolving rapidly. Second, the range of applications to day-to-day practice is broad. Third, the need for solutions will heighten scrutiny of educational outcomes and assessments, and, fourth, the policy environment and other exogenous constraints are dynamic. We explore each of these categories below.

Subject Matter Complexity Dealing with subject matter complexity depends on whether the complexity is determined by disciplinary depth or multidisciplinary breadth, or, as in many cases, by both. In addition, disciplinary considerations interact with one another, leading to a network of linkages that requires investigation outside of disciplinary-specific epistemological norms. What we see emerging in our interactions with executive education clients with transdisciplinary concerns, as well as with our own degree program students, are interests in challenges that combine complex fundamentals with socioeconomic and operational impacts. We have seen this explicitly in partners' concerns with, for example, the absorption of artificial intelligence into decision-making, or on the quality and accuracy of data sources, or on the socioeconomic barriers to pandemic response. Finding the right approach to an executive education intensive, let alone multi-year undergraduate or graduate degree programs, requires innovations in curriculum, pedagogy, and program management.

Building the curricular elements requires input from a larger number of faculty representing an appropriately larger number of disciplines, engaging in comprehensive discussions of interconnections, syllabus ladders, and co-teaching. In addition, thought leaders within the faculty need to construct intellectual threads that account for the intersectionality of the core topical challenges and the ways in which the

individual research contributions of the disciplines are combined into a more seamless and holistic framework.

Most university faculty are obviously comfortable teaching within their area of specialty, but teaching in a transdisciplinary curriculum places an additional burden on more specialized faculty by requiring them to identify and speak to the intersections with the other disciplines and develop curricular segments devoted to systems thinking or convergence. Identifying faculty for whom this comes naturally or training faculty to address the overall curriculum present particular challenges. Faculty incentives and instructional support need to be enhanced to address these new considerations.

While the number of faculty engaging in curricular buildout might approach the scale of a medium-sized academic department, executive education managers should develop a smaller leadership group that will be familiar and comfortable with new approaches and bringing new research from fields outside their own into the classroom.

Innovations in program management include managing large numbers of faculty from several departments and schools, distributing resources and revenue, devising a marketing strategy, implementing *ex post* assessments, and dealing with students who themselves may not have been exposed to a multidisciplinary, let alone a transdisciplinary, instructional journey. The marketing strategy has to point to the uniqueness of such an approach, while course assessments need to explore how students have moved beyond their own disciplinary backgrounds.

Broad Applications to Practice Potential students approach executive education to broaden and deepen their skill set, and to learn about new dynamics influencing their current or anticipated job responsibilities and their institution's growth and performance. Transdisciplinary topics, however, do not map immediately into job tasks back at their desks. Curriculum design has to navigate the gulf between individual needs for immediately usable tools and a new work environment informed by the breadth and intersectionality of a complex topic.

Part of the challenge is redefining the parameters of practice, which should be one of the objectives of the curriculum. If well-articulated, such redefinition could contribute to altering the strategic direction of the students' home institution. This will require a thorough understanding of the existing practices in the student cohort, and the expectations of the enterprise with respect to the outcomes of the education experience. The curriculum needs to establish a practice trajectory that provides the students with goals and benchmarks through which they can apply their learnings to their own careers and through which their company or enterprise can leverage their new capabilities and understanding as part of its strategic planning. This could be as simple as learning to access new datasets or as complex as running integrated models and developing scenarios with enlarged sets of inputs.

We have found that executive education students often arrive with an expectation that they will be taught how to use new tools designed to amplify their basic job responsibilities and performance within the context of current practice. This

expectation should be discussed during the introduction to the curriculum and should be expanded in concert with new practice parameters.

Solutions Orientation Somewhat in parallel to practice issues, interest in complex subject matter arises because of the need for the home institutions to innovate and implement solutions as part of their business strategies. Depending on the domain, solutions might range from technological innovation to operational changes to new modes of financing to responding to new consumer markets. It might be necessary to cover the spectrum of solutions in an open enrollment course, or for a particular partner, such as a consulting group, with a correspondingly wide range of practice areas. In any case, just as the subject matter is dynamic and changing rapidly, the practicality and sustainability of solutions can also shift as new knowledge emerges and is translated into applications.

We have found it useful to outline the case for solutions with respect to adoption timescales, that is, what might be practical in the near-, intermediate, and long-term with respect to a particular area of business practice. Further, we include available research on the impact of various solutions on achieving desired outcomes with respect to the underlying challenges, both for the individual enterprise and for more global objectives.

Policy Dynamics Perhaps the most complicating factor in developing executive education curricula for complex subject matter, is the way in which exogenous factors—including policy, political, legal, and regulatory environments—affect the operations of educational partners. The interplay between rising public awareness of societal challenges and concomitant political responses can provoke rapid and unpredictable changes in policy, regulatory and the legal frameworks in which institutions and enterprises operate. This affects curricula by imposing a strong requirement that the rationale behind these changes, their impact on the sectors in which the students and their institutions are operating, and the technical and tactical responses to them must be part of the curriculum and addressed by the instructors. Moreover, partners sending students to executive education increasingly operate in multinational, global markets, which requires close examination of differences in regional policies, cross-border issues, and the geographic variation in the underpinning phenomena driving the whole educational effort.

Particularly for subject matter underlain by highly technical research in science and engineering, core scientific and engineering faculty might not be able to address the policy dynamics inherent in the grander challenge. Thus, adding these elements to the curriculum requires input from experts in business, law, and policy, for example. Including these faculty from the start of the consultation and design process provides better integration and more approachable and relevant learning objectives for the students.

9.3 Case Study: Climate and Sustainability Executive Education

We provide here a detailed examination of how these general principles are applied in the case of building an executive education curriculum that addresses climate change and sustainability. Our group has had extraordinarily detailed discussions with a number of external partners from diverse business sectors and with differing strategic goals. However, we have found that the general principles of design enumerated above can be successfully applied across different target audiences.

Columbia University has a history of research and education in fundamental Earth and environmental science and engineering. Our professional schools of business, international and public affairs, law, public health and our School of Professional Studies each host institutes and research centers as well as degree programs focused on interactions between human society and the natural world, or to be more precise, climate, sustainability, and sustainable development. As global warming and climate change rose as priorities in the public sphere, university leadership worked with senior faculty to design a new professional school that would provide an administrative focus on transdisciplinary scholarship and degree programs, and accelerate research across the university's academic departments and institutes. The culmination of this work is the Columbia Climate School, a new professional school having among other responsibilities a central coordinating position among the other professional schools and academic departments. Other leading academic institutions have created schools and institutes with similar goals and functions.

We do not argue here about the existence of global warming and climate change, other than to observe that the scientific evidence is sound and the risks are becoming apparent. Instead, we point to the ways in which governments, consumers, and private enterprise are motivated to reduce the physical impacts of climate change and find solutions that are practical, achievable, and economically and socially sustainable. The uncertainties about how society should respond are creating new demands on university research and degree and nondegree education alike as well as opening up new avenues of university-industry engagements. Climate change and accompanying issues in sustainability can serve as a type example of the need for innovative, transdisciplinary design. Enlarging executive education program offerings is one component of "innovation at the interface."

The institutional innovation required for the formulation and implementation of a new school speaks to the increasing need for universities to present new models of scholarship and stakeholder engagements that place universities firmly in the public square in order to help generate solutions to complex and pressing challenges. This is an imperative not only in the climate and sustainability space, but across the complex challenges that we have characterized as driving the need for new degree programs and related executive education curricula. Columbia calls this its "Fourth Purpose," a responsibility added to the traditional functions of scholarship, teaching and disciplinary service. Similarly, Malkov et al. (2024) have written of "Fourth

Generation Universities” having an enlarged social responsibility to accelerate the translation of research and participate in the innovation required to meet these challenges.

Whether fourth purpose or fourth generation, climate and sustainability present such challenges. Rapid developments in the core basic research disciplines, new transdisciplinary research agendas with systems oriented methodologies, and the rapid rise in public awareness of climate impacts in particular illuminate the need for innovative responses in executive education. Potential executive education cohorts are concerned with rapid and politically oscillatory changes in policies and regulations affecting enterprise behavior, and are seeing additional pressure from their clients. Corporate leadership often sees an advantage of being “first” among their peers in responding to climate and improving corporate sustainability, let alone participating in or providing the innovation needed for solutions.

Thus the curricular design of executive education courses, whether delivered intensively or on a semester-long schedule, can be derived from graduate and professional degree programs undergoing concurrent, innovative fourth-purpose or fourth-generation shifts. Similarly, engaging motivated faculty in developing these executive education curricula provides a feedback mechanism that will bring external relevance to the degree programs offered to undergraduate and graduate students enrolled in the university.

9.3.1 Learning Objectives for Climate and Sustainability Executive Education

Columbia’s undergraduate, graduate, and professional degree programs with climate- or sustainability-centered curricula provide the raw material for constructing a comprehensive climate and sustainability executive education program and serve as a marker of faculty research expertise and teaching interest. We begin the executive education curriculum design exercise by enumerating learning objectives and core competencies derived from the degree programs and tuning them to the needs of potential executive student cohorts. A small faculty committee works with the executive education leadership team and a small group of external advisors provides additional vetting.

An underlying premise is that even if executive education cohorts agree *a priori* that dealing with climate and sustainability are critical to their institution’s resilience and viability, it is still important to review how basic research has contributed to the view of climate and sustainability as a set of operational, socioeconomic and even humanistic challenges which should be recognized and resolved by all of society, let alone by individual enterprises. That basic research evolves and continues to provide input, constraints, and direction to governments and international policy bodies signifies the dynamic nature of the challenge. The fluidity of the challenge, especially with respect to predictive uncertainties and exogenous constraints, adds

a level of complexity to the curriculum and indicates the importance of experiential content (case studies and exercises) and even co-generation of syllabi with external partners.

We illustrate this example with a subset of climate and sustainability learning objectives. Each illustrates how complex transdisciplinary topics can be distilled into a useful programmatic outline. The degree of detail presented here is meant to be illustrative of a one-week, 40-hour intensive course.

9.3.2 Climate Science and Climate Impacts

We need to take the cohort beyond the so-called “popular understanding” of climate science, which can run into hyperbole and doom-saying, and deal with denialism in some quarters. We start by emphasizing data-based conclusions about warming temperatures and explain nuances in the geographic distribution of sea-level rise. The syllabus introduces concepts such as emergent versus extreme events, climate thresholds and irreversible changes, the attribution of extreme weather to background climate change, and the rising importance of correlated extremes such as extreme heat and humidity.

Climate modeling and the tools of climate prediction are frontier research areas, with recent advances in the application of artificial intelligence to computational research helping to unravel some of the complexity and uncertainty in model results. We emphasize the role of uncertainties and how data analysis and modeling are used for probabilistic forecasts and scenario-building.

We underplate the discussions by remarking on the importance of time and spatial scales, from short-term to long-term forecasts and from global to local impacts. The notion here is that business decisions of a particular scope and scale can be supported by research of similar scope and scale, reflecting the spectrum of decisions typically driven by different risks and opportunities.

We introduce climate risk as a decision metric and begin to link fundamental impact research (How much sea-level rise and when? Forecasts for the next drought? The seasonal forecasts for hurricanes and tornados?) to the operational decisions we discuss later in the course. Depending on the specific interests of the cohort, we might survey the impacts of climate change on different economic sectors, such as manufacturing, real property, agriculture, public health, consumer products, insurance, and labor. More generally, we discuss physical risks and transition risks and introduce the concepts of risk identification and management that will be discussed in a later session. The focus on climate risk begins to relate the basic science of impact analysis with enlarging the spectrum of risks that should be considered by business. Key to this is characterizing risks as physical, economic, reputational, legal and preparing for later sessions exploring these perspectives.

9.3.3 Domestic and International Policy Environment

Our industry partners work within domestic and global policy environments that might incentivize climate-sensitive actions. Policy can also be volatile, changing rapidly with political turnover. We work with our faculty in law and public policy to ensure that we present the most up-to-date perspectives on climate policy and law.

This includes global policy, which will be of interest to companies having global presence and reach, or which depend on globally ranging supply chains and markets. The United Nations Framework Convention on Climate Change (UNFCCC) has established the annual Conference of the Parties (COP) as the decision-making body for setting emissions reduction parameters for parties to the Convention, which in turn are meant to influence national policies and commitments. Companies with global operations may find themselves having to operate under a complex weave of national and international regulations and cross-border adjustments. We have found, somewhat surprisingly, that the interplay among global policy, national commitments and business strategies, and the ways in which these are aggregated to approach climate goals, is not well understood by many of our partner cohorts. Working through this example takes up a substantial portion of this session.

Separate but perhaps no less influential are the demands of clients and the public for changes in corporate behavior and the leadership response. After scoping needs with partner liaisons, we develop exercises for the cohort to work through applications in their particular area of practice.

Depending on the degree of familiarity, we outline the UNFCCC process, summarize recent reports and review the discussions and potential obligations arising from the most recent COP and expectations for the next one. We also review recent domestic legislation and executive actions, and explore differences with European, Asian, and the Global South.

9.3.4 Energy Transition and Decarbonization Pathways

The “transition to net-zero or net-negative” and “decarbonization pathways” and “decarbonized economies” are some popular terms of public discourse that serve as delimiters of corporate action and policy environments. This language explicitly admits that achieving solutions to the climate problem and manifesting sustainability is a journey in research, adoption of green technology, and changes in policy and corporate behaviors, and not a single watershed moment. This allows us to introduce scenario modeling that factors changes in socioeconomic behavior into climate forecasts, so that our partners see their responses as a series of discrete decisions over time.

We focus on how these popular terms are underlain by solid science with a realistic assessment of how green technologies might be adopted, enabling discussions of scenarios and the fit of corporate strategy to them.

9.3.5 Mitigation, Adaptation, Resilience, and Vulnerability

Some terms in popular, informal use have specific meanings in policy and regulatory documents and require more precise definitions in the curriculum. In the climate and sustainability world, these include mitigation, adaptation, resilience, and vulnerability. We devote a session to policy-specific definitions of terms that would be used in developing strategic objectives. Often this involves classifying particular actions according to the terms appearing in scientific and policy documents. For example, climate “mitigation” includes those actions that serve to reduce greenhouse gas concentrations in the atmosphere, including, for example, reducing emissions or capturing emissions downstream of the manufacturing or energy generation process. “Adaptation” concerns the actions that could reduce future damage. These sessions ensure that instructors and students share a lexicon in practice.

Apart from their role in specific policy documents, these terms provide a way to introduce socioeconomic concepts such as disproportionate impacts, climate and environmental justice, emerging market financing, and climate migration and changing demographics.

9.3.6 Climate Tech and Innovation

There is significant academic and commercial research on renewable energy technology, battery electrochemistry, carbon dioxide capture and sequestration, green mining, and green manufacturing. There are also rapidly emerging and evolving national industrial policies meant to strengthen national standing while developing green economies and accelerating the adoption of new technologies. Having worked through the concepts of climate risk, we survey progress in green technology as an introduction to opportunities for corporate research and development and investment and commercialization, and the prospects for different technologies to contribute to the energy transition. We also identify potentially disruptive technologies that might change the arc of decarbonization pathways and climate scenarios. We complement these sessions by bringing in university tech transfer officials and entrepreneurial advisors and discuss real-world examples of industry-sponsored university research, technology patenting and licensing, and venture investing.

9.3.7 Climate Finance

“Climate finance” generally describes how government, philanthropic, and private assets might be reallocated to encourage progress toward a sustainable future. The topic is itself complex, encompassing for example the description of specific financial instruments to geopolitical issues concerning green investments in developing

countries. By broadening the spectrum of investment risk to include climate-related risk, the curriculum can lead to changes in the decision cultures of financial institutions in ways that might accelerate the transition to more decarbonized energy, manufacturing, agricultural and transportation sectors, for example.

The curriculum can link climate finance to the emergence of new technologies by discussing the readiness of new climate tech to enter the markets, the potential impact of new tech on climate trends, and the relation to investment time scales. Case studies presented by university tech transfer officers provide an excellent template for these kinds of discussions.

9.3.8 Other Topics

The curriculum might cover additional topics of interest to different cohorts. Industrial ecology and circular economies, disproportionate climate impacts in certain communities, justice and fairness in climate investment, climate-forced migration and geopolitical impact and climate communications have all been of interest, depending on the audience. Each of these topics can generate specific learning objectives which can be linked to the core objectives described above.

9.4 Challenges and Opportunities: Customized Curriculum

Coordinating executive education and university-industry relationships when the relationships are assembled around leading societal challenges rather than simply-articulated workforce upskilling presents challenges to both universities and their partners. The spectrum of disciplines and faculty expertise specifically necessitated by the transdisciplinary nature of the problem requires faculty collaborations both in research and teaching, as well as administrative incentives that encourage intellectual crosstalk. Whereas some traditional executive education courses might involve just a handful of faculty contributors, as in the example above, a transdisciplinary curriculum could in theory require dozens. In practice, assembling an executive education team around such complex subject matter needs to reflect broader university initiatives that encourage faculty collaborations across traditional schools and departments.

The perspectives of university partners will range from the specific applications of new knowledge to simply developing a better understanding of unknowns and uncertainties. This often manifests as a query to receive training on risks and unforeseen or exotic operational externalities, and the strategic opportunities that might arise from understanding the need for solutions to these types of problems. Moreover, we have come across cases where the discussion has to be defined, that

is, we have needed to work with some partners to develop questions that they should be asking, let alone the questions that we could ask them. Some partners may not have had the focused internal discussions that would set the stage for engagement with a comprehensive executive education program.

This necessitates a dialog with partners focused on balancing a curriculum based solely on faculty expertise alongside curricular topics articulated by student cohorts and their companies. Such co-generation of topics has the advantage of self-marketing to a particular partner but presents a challenge to make the content broadly applicable to the needs and expectations of a range of partners.

We are brought back to the notion that executive education for the complex challenges facing society requires input from a multitude of faculty and partners, some of whom may not anticipate the questions that should be addressed. The need for both breadth and depth translates into a design challenge for executive education, but also provides feedback for new approaches to undergraduate and graduate pedagogy.

The expansion of traditional ideas of executive education and industry partnerships having a knowledge-building purpose presents opportunities that filter back to undergraduate and graduate curricula. First, the need to explore the totality of such complex problems presents an opportunity to develop a truly transdisciplinary course ladder, with disciplinary linkages explicitly reflected in individual syllabi. Second, a transdisciplinary pedagogic approach requires the involvement of multiple faculty across departments, institutes, and centers, which can lead to collaborations that can in turn lead to new research trajectories. Faculty co-teaching across disciplines often find ways to pursue interesting new research based on their work in the course.

We categorize the challenges as (1) faculty participation; (2) coherent curricular design; (3) matching curriculum to partner needs; (4) incorporating rapid developments in research and external factors affecting partner strategies and operations; and (5) marketing, assessments, execution, and links to deeper engagement.

9.4.1 Faculty Participation

The transdisciplinary structure of grand societal challenges presents university leadership with the challenge of incentivizing participation of traditionally siloed faculty. The challenge is broader than simply convening a diverse executive education faculty, and can lead to new initiatives across schools and departments. Typical tools include internal investment by central administration, adjustments to administrative and financial relationships among departments and schools, and identifying peer faculty leaders with the ability to build new approaches to research and education.

9.4.2 *Coherent Curriculum Design*

Curriculum designers are faced with designing a holistic, transdisciplinary treatment of subject matter that might otherwise be covered by hundreds of semester-long graduate and undergraduate courses in the degree programs. In the case of the climate and sustainability example at Columbia, these number more than a hundred in the masters programs alone.

The challenge falls to faculty leadership to find one or more thematic lines through the individual course syllabi and gleaning content adhering to these themes. Using the climate and sustainability example, we have found that the transition to a “decarbonized” energy sector is one such theme that can be delivered over the span of a few day-long sessions. We have used a “sector” designation, such as agriculture or transportation or finance, to build other thematic approaches, and have combined them into several week-long executive education programs. However, each program begins with a customized survey of the underlying science.

The disciplines contributing to an understanding of complex societal problems often evolve rapidly, as they themselves undergo transformative change associated with highly collaborative work driven by the imperative to work on these questions. Faculty endeavor to channel rapidly changing frontier knowledge into their individual degree program syllabi, especially at the graduate level. Linking these individual advances to the executive education curriculum is another challenge, as well as an opportunity to market the program as adjacent to leading research. Executive education leadership can build a dynamic administrative approach, usually through actively engaged curriculum review committees, ensuring rapid adoption of frontier concepts. University partners facing their own internal and exogenous imperatives for rapid change will find a certain synchronicity with this approach.

9.4.3 *Matching Partner Needs*

Often, a faculty-designed curriculum based on faculty interpretations of what students need to know—the traditional university educational model of syllabus development—will meet the needs of many if not most executive education clients. On the other hand, designing a curriculum to match the specific needs of a particular partner can be time-consuming and sometimes frustrating. The difficulties include: (1) effectiveness of a client liaison; (2) having ill-defined or undefined discussion topics; (3) client interests in level of applicability versus elevating staff’s background understanding; and (4) distinguishing between executive education and consulting services. These difficulties adhere to any bespoke curriculum, but are particularly relevant when addressing the new content associated with complex societal challenges.

First, the client’s liaison to the curriculum design team should be knowledgeable about the issues, of course, and should have a clear connection to client leadership

and the strategic rationale behind the need for a bespoke curriculum. This rationale should be held up against potential course and program syllabi and curricula so that rigorous and useful outcomes can be achieved. Convergence on an outcome the client considers useful requires iteration. If not successful, both sides have to agree to a compromise. Friction and frustration might ensue.

Second, even with a knowledgeable and strategically connected client liaison, the client may have only vague articulation of the topics that need to be addressed, or may, in the opinion of program leaders, have a formulation that is a poor match for faculty expertise. Our experience suggests that client interests tend to be at a level of granularity different from what faculty normally consider to be appropriate topics. Adjustments need to be made through design meetings.

Third, the first two concerns typically lead to a discussion of particular tools (datasets, analytical methods) that the cohort can bring back to the enterprise to use on a daily basis. This gets at the question of what it means to “change the decision culture” of an enterprise to more effectively deal with the issues associated with the complex problem. In our discussions, we cast this as a matter of scale in both time and place. The elements of the disciplines that contribute to changing the culture need to match the scale of decisions in both space and time. We make this especially clear when we talk about climate and weather predictability, for example, or the range of investments in solutions (mitigation and adaptation) that have time scales associated with return on investments.

Finally, expectations must be managed when engaging in client discussions about bespoke programs lest the arrangement drift into what might be better served by externally funded research or even private consulting.

9.4.4 Incorporating Rapid Changes in Disciplinary and Transdisciplinary Research

University executive education holds an advantage if its relationship to frontier university scholarship results in timely translation of new knowledge into executive education curricula. This is especially true as universities encourage research on complex societal problems. Typically, though, instructors will need to engage with student cohorts to emphasize transdisciplinary learning, that is, leading students to think beyond their disciplinary expertise to look for connective links to other fields. This puts extra demands on the faculty to adopt a more open pedagogic approach. Curriculum review committees with active connections to the research enterprise and pedagogic support offices should be charged with frequent programmatic reviews.

9.4.5 Marketing, Assessments, and Deeper Engagements

The growing case for offering comprehensive curricula addressing complex societal needs can align with the university's own perception of its purpose outside of traditional teaching, research, and disciplinary service. The argument here is that developing transdisciplinary literacy through executive education begins to reorient partner enterprises toward managing their risk and thus reducing costs on society, orienting their enterprise toward sustainability, and working to develop solutions within their operational or sector domains. This should be part of the executive education program's marketing strategy.

In parallel, *ex poste* assessments need to be bi-directional. Traditional student evaluations are useful, but there needs to be stringent assessment by the client as to whether the clients' goals have been met. This sort of feedback has been essential to us as we map emerging university initiatives focused on transdisciplinary research into custom and standardized executive education curricula.

Finally, by introducing clients to new concepts through executive training—especially when there are preliminary discussions of what the university can bring to a more formal partnership—executive education faculty engagement with external partners may lead to student internships, external funding for research and other outcomes broadly aligned with university development, research and degree education.

9.5 Bespoke Implementation: Meeting Specific Learning Objectives for a Multi-Sector Firm

We illustrate our approach with a custom curriculum for a five-day intensive delivered by about two dozen faculty through lectures, panels, discussion sessions, keynote addresses, and accompanying exercises. The curriculum was developed in response to a specific request from a firm whose clients range broadly across sectors. The firm sought to provide its most highly ranked client-facing staff with the insights and tools they would need to interact with their clients across all practice areas. The firm's clients include private and public sector organizations, philanthropies, and nongovernmental organizations, with both domestic and international operations. Initial discussions with the company's senior managers confirmed the need for a training arc grounded in basic and applied research and attention to the global and domestic policy environments in which their clients needed to operate. We emerged from these preparatory discussions with a mandate to deliver on themes of climate risks and opportunities; greenhouse gas mitigation and climate adaptation, institutional and jurisdictional resiliency, and the terms and characteristics of a sector-by-sector transition to a decarbonized world.

Our curriculum adheres to the learning objectives we have described. We aimed to provide our student participants with a comprehensive introduction to the

elements of climate science that are drivers of policy, regulation, customer needs, innovation, and how this will influence enterprise practice centered on providing solutions to society and improving enterprise resilience and viability.

Table 9.1 displays the co-developed curriculum outline. The topics covered in the program amount to about one full three-credit, transdisciplinary course, and the material is presented at the graduate level. The exercises are built around team collaborations and are designed to reinforce the intersections across disciplines and sectors, with an emphasis on the impacts of complexity and uncertainty on a firm’s decision culture.

Table 9.1 Curriculum

Day 1: Framing the Challenge: Climate Change, Climate Impacts, and the Transition to a Decarbonized (1.5°) World
Basic climate science: What do we know, and where are the frontiers?
Climate impacts: Defining physical and transition risks across sectors
Case study: Modeling offshore wind
Day 2: The Energy Transition and Decarbonization
Sector overview: Power, transport, built infrastructure
Critical minerals and supply chains
New tech for decarbonization: Batteries, alternative fuels, concrete, and steel
Carbon capture, utilization, sequestration
Case study: Modern geopolitics of the energy transition
Day 3: Adaptation and Resilience
Transitions in the food system: Nutrition, livelihoods, and sustainability
Financing the transition: Sustainable investing, green bonds, and carbon markets
Financial instruments for the energy transition in emerging markets
Case study: New York City: Converging risk in the era of climate change
Day 4: Special Topics (Organized After Discussion with Corporate Liaison)
Circular economy
Biodiversity
Macro and corporate perspectives on climate risks and opportunities
Company disclosures
Corporate challenges (group activity)
Keynote: Macro and micro economics of the transition
Day 5: Recap and Summaries
Overall recap (conveners)
Industry intersections: Discussion and exercise (application to company clients)
Climate action moving forward
Closing summary and farewell

9.5.1 Initial Assessments

Versions of this curriculum, with fine adjustments depending on client needs, have been well received by other partners. Participants indicate that they have learned more about the climate system and the calculation of climate risks than they would have gotten through other means. They gained insights into actions taken by nations in response to global agreements and the ways in which individual companies might align with national obligations.

Descriptions of new technologies and time scales for adoption generally receive the best ratings. Similarly, high-level discussions of domestic and global policies generate substantial interest, even if not directly relatable to individual work responsibilities.

Our curriculum emphasizes the novel aspects of learning about and dealing with complex societal challenges such as climate and sustainability, as well as their applications to business practice. Nevertheless, we still receive comments about the need for specific applications to existing modes of work. This continues to be a discussion point for our faculty.

Most of our faculty have become adept at dealing with the diverse interests and preparation of student cohorts through their work in our degree programs. The diversity in our executive education programs is even more granular, with many students having a particularly narrow view of applications to their own work responsibilities. This diversity presents both a pedagogic and design challenge. We have found that group exercises that require multidisciplinary teams along with pre-reading lists that promote a minimum level of understanding alleviate part of this challenge. We continue to work on this aspect of curriculum design.

Finally, forty hours of instruction delivered over five days can exhaust attention spans. In response to assessments, we have scheduled more discussion breaks, group exercises, and afternoon and evening social events.

9.6 Takeaways and Future Strategies

For universities that promote engagements with external partners to analyze and develop solutions to broad societal challenges, comprehensive executive education programs can be a tool to explore common interests with those organizations.

The manner in which programs are delivered can resemble traditional university coursework or can be constructed around more interactive workshop models, panel discussions, and keynote speaking engagements. Done well, a combination of modalities makes for an effective and stimulating experience for participants and strengthens the relationship between the university and its partners. By introducing important participant cohorts to individual faculty and their research, executive education advances university-industry partnerships.

Emerging and difficult societal challenges are characterized by their complexity and intersections among a broad spectrum of disciplines. Accounting for these intersections leads to the need for a set of intellectual threads through the disciplines, leading to coherent research and degree and nondegree curricula, what has been termed “transdisciplinary.” Moreover, complex societal challenges often drive a market for executive education experiences. Universities are in a solid position to offer a response.

Transdisciplinary challenges impact potential university partners who may turn to faculty expertise for advice and training. Executive education, long focused on subjects relevant to conventional practice, can evolve to help partners adapt and prosper. We offer these takeaways:

1. Training for transdisciplinary challenges is an emerging market which differs from what most traditional executive education programs can offer.
2. This aligns with increasing university efforts to address these challenges through advances in research and education.
3. The demands of a transdisciplinary program require innovation in academic structure, faculty collaboration, curricular design, and classroom pedagogy.
4. Research addressing complex societal challenges evolves rapidly, necessitating a dynamic approach to administration and educational programs,
5. The importance of addressing these challenges and the market requesting interaction can amplify university-industry partnerships, not just in education, but in research as well.

Universities are adapting to expand their footprints beyond traditional areas of research, education, and professional service to include more substantial interactions with broader society in developing solutions to our most pressing challenges. Conferences and workshops should be convened to explore the best practices across the range of university administration of schools and disciplinary departments. Pertinent curricula could be published and distributed to serve as baseline resources for degree and nondegree education.

Reference

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